

PRAJWAL V

Bengaluru, India | +91-7411767592 | vprajwal2204@gmail.com | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Cybersecurity Analyst and Software Engineer specializing in Security Operations Center (SOC) tooling, automated threat triage, and digital forensics. Experienced in building custom defensive applications, optimizing log analysis pipelines, and analyzing low-level system security. Adept at transforming raw network telemetry and host logs into actionable security alerts to accelerate incident response.

TECHNICAL SKILLS

- **Security Operations & DFIR:** Digital Forensics & Incident Response (DFIR), Host-Level Forensic Analysis, Threat Hunting, Vulnerability Management.
 - **Network & App Defense:** Intrusion Detection (Suricata, Snort), Traffic Analysis (Wireshark), OWASP Top 10, Static Malware Triage.
 - **Languages & Scripting:** Python (Security Automation & Parsing), Go (Systems Security & Isolation), Bash, TypeScript, C/C++.
 - **DevSecOps & Tools:** SIEM (Splunk / ELK), Linux (Kali), Docker, Git, Burp Suite, Nmap.
-

HANDS-ON SECURITY EXPERIENCE

Frostbyte | Digital Forensics Framework | *Python* | [Repository](#)

- **Developed** a self-contained digital forensics engine to parse and decode binary Windows artifacts natively without relying on OS-dependent tools.
- **Built** native parsers for volatile host artifacts to extract system execution timelines, registry state changes, and persistent Indicators of Compromise (IOCs).
- **Optimized** disk log processing modules, creating a structured environment for incident responders to automate host-level threat scanning across complex filesystem structures.

Nexus | Next-Gen SOC Platform | *Python, REST APIs* | [Repository](#)

- **Engineered** a real-time SOC monitoring platform focused on tracking system behavior, log aggregation, and automated incident mapping.
- **Orchestrated** continuous event-processing pipelines to ingest host and network logs, applying automated correlation rules to handle simulated peak loads of 500+ events/sec.
- **Designed** a centralized security dashboard that aggregates fragmented telemetry, reducing duplicate alerts and drastically improving analyst visibility during triage.

AegisTriage | Static Binary Triage Platform | *Python* | [Repository](#)

- **Spearheaded** a modular static analysis tool designed to digest untrusted file samples, extract cryptographic hashes, and classify files using structured signatures.
- **Programmed** safe extraction routines to pull embedded strings, calculate file section entropy, and identify suspicious import tables without executing the file.
- **Accelerated** initial triage times by automating the detection of common packer signatures, passing normalized risk scores directly to incident queues.

docksmith | Custom Container Runtime & Builder | *Go* | [Repository](#)

- **Authored** a minimalist container image builder and runtime from scratch in Go to deeply analyze the low-level security boundaries of containerized systems.
- **Enforced** strict process isolation using native Linux kernel primitives (**Namespaces, cgroups, chroot**), implementing secure PID, filesystem, and user boundaries.
- **Applied** system-level engineering research to build defensive guidelines for identifying container escapes and privilege escalation vulnerabilities in cloud environments.

ACADEMIC CAPSTONE PROJECT

PQC Federated Anomaly Detection for Connected Vehicles (V2V) | *Docker, Python, TensorFlow, Flower*

- **Co-engineered** a containerized, privacy-preserving Federated Learning framework that trains a hybrid **1D-CNN + GRU** model across decentralized vehicle nodes to detect malicious Basic Safety Message (BSM) injection.
- **Implemented** a dynamic, threshold-based network gateway control that shifts the communication stack from standard ECC to quantum-safe protocols (**Kyber + Dilithium or Falcon**) immediately upon runtime anomaly detection.
- **Utilized** the Flower framework and FedProx optimization to manage non-IID datasets, integrating Byzantine-resilient aggregation to isolate malicious nodes and mitigate model poisoning attacks.

SECURITY LABS & PLATFORMS

- **Capture The Flag (CTF):** Actively compete in global Capture The Flag challenges, specializing in solving forensic timeline reconstruction, binary static analysis, and network packet analysis vectors.
 - **Defensive Labs:** Completed 40+ hands-on blue-team and DFIR labs across **TryHackMe** and **Security Blue Team** platforms covering enterprise incident response, SIEM log analysis, and Active Directory defense.
-

SECURITY RESEARCH & LAB REPORTS

- **Network Infiltration & IDS Bypass:** Researched and documented technical methods for bypassing stateful Suricata setups using packet fragmentation and timing-based obfuscation.
 - **Web Application Penetration Testing:** Conducted security assessments across deliberately vulnerable lab environments (**DVWA**), configuring 50+ custom rules targeting high-risk web application attacks mapped directly to OWASP Top 10 mitigation strategies.
-

CERTIFICATIONS

- **Ethical Hacker** – Cisco Networking Academy (2026)
 - **Introduction to Penetration Testing** – Security Blue Team (2026)
 - **Cybersecurity Course** – Tech Mahindra Foundation / Skill India (2026)
 - **Professional Development:** Currently preparing for eLearnSecurity Junior Penetration Tester (eJPT) and CompTIA Security+
-

EDUCATION

Bachelor of Technology in Computer Science and Engineering

PES University, Karnataka | Expected Graduation: May 2027